VICTORY 2027

Water Resources

LECTURE- 2 | Geography

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Topics To Be Covered

1. Irrigation → सिंचाई
2. Importance and methods
3.
4.



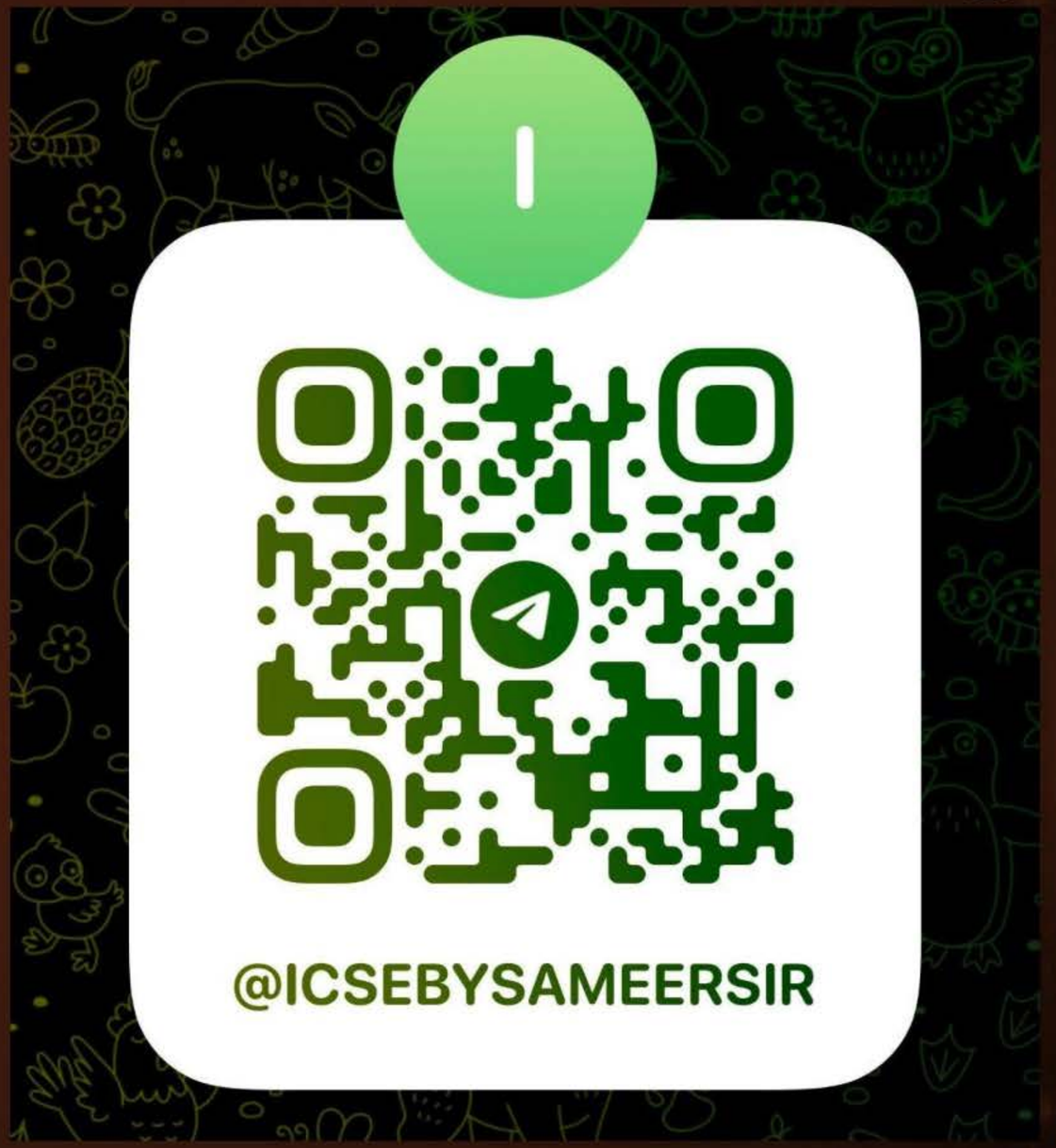
Water Resources

- *Sources (Surface water and ground water).*
- *Need for conservation*
- *Rain water harvesting and its importance.*
- *Irrigation: Importance and methods – wells, tube well, tanks, canal, drip and sprinkler irrigation.*


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11. Despite receiving abundant rainfall during the monsoon, India faces acute water scarcity in many regions. What is the main geographical reason for this?

- A) The rainfall is highly erratic, seasonal, and unevenly distributed.
- B) Indian soils cannot absorb any rainwater.
- C) All rainwater instantly evaporates due to the tropical climate.
- D) India lacks surface water bodies completely.

13. What is an 'aquifer'?

- A) A man-made reservoir behind a multi-purpose dam
- B) A natural underground layer of porous, water-bearing rock or gravel
- C) A concrete-lined irrigation canal
- D) A traditional water storage tank in South India

Means Of Irrigation

- Based on the availability of surface water (rivers and streams) and groundwater, relief, soils and climatic conditions, different means of irrigation are utilised in India.

- For example, in the crystalline soil areas of Deccan Plateau, tanks form the major source of irrigation, whereas the alluvial tracts of the Gangetic and Coastal plains are best suited for canals and well irrigation.

Means Of Irrigation

old

- The conventional means of irrigation include wells, tanks, and inundation canals.

Too much of rain water

- Irrigation is usually done manually, where a farmer himself pulls out water from wells, tanks and canals or uses animals to draw water.

- The modern means of irrigation include tubewells, perennial canals, drip irrigation, sprinklers etc.

Wells

- A well is a small hole dug in the surface of the earth to obtain water from the water table for irrigation and other purposes.
- Wells are usually of two types: **Lined wells and Unlined wells.**
- **Lined wells are man-made; they are artificially constructed by digging into the ground and supporting the walls with bricks in a circular pattern.**
- **Unlined wells are formed by different natural phenomena and are normally not protected by lining of their walls.**





Wells

- Well irrigation is a cheap, simple and dependable source of irrigation.
- It is practised in areas where plenty of groundwater is available and the depth of the water table does not exceed 15m.
- **Area:** The areas where well irrigation is practised include Punjab, Uttar Pradesh, Goa, Haryana, Bihar, Rajasthan, Gujarat, Maharashtra, Madhya Pradesh, Andhra Pradesh, Karnataka and Tamil Nadu.

Wells

The water from the wells is lifted for irrigation in a number of ways:

(i) **Persian wheel method: A Persian wheel is a traditional, animal-powered mechanical device used to lift water from open wells or streams for crop irrigation.**

- It is used mainly in Punjab, Haryana and Western Rajasthan.
- It is a partly submerged vertical wheel with buckets attached to the rim. As the wheel is turned by animals rotating a geared horizontal wheel, the buckets are filled and emptied into a ~~trough~~ above, which carries the water to crop-sown fields.

↳ Container





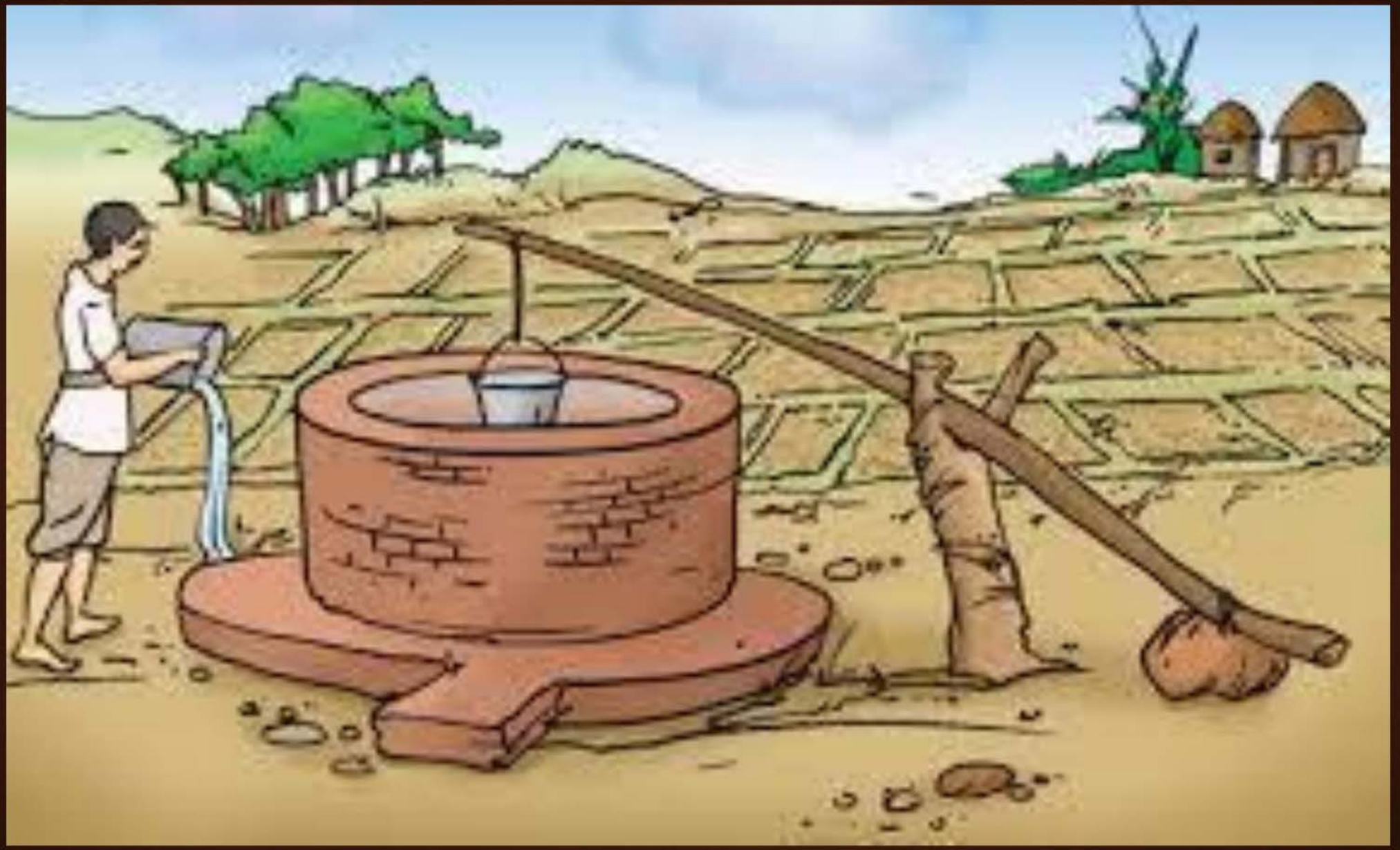
Persian Wheel

Wells

(ii) **Lever method:** It is used in Bihar and Andhra Pradesh. It is an economical and efficient method of lifting water from shallow wells to a height of 3.5 to 4.5m.

→ Dhekli

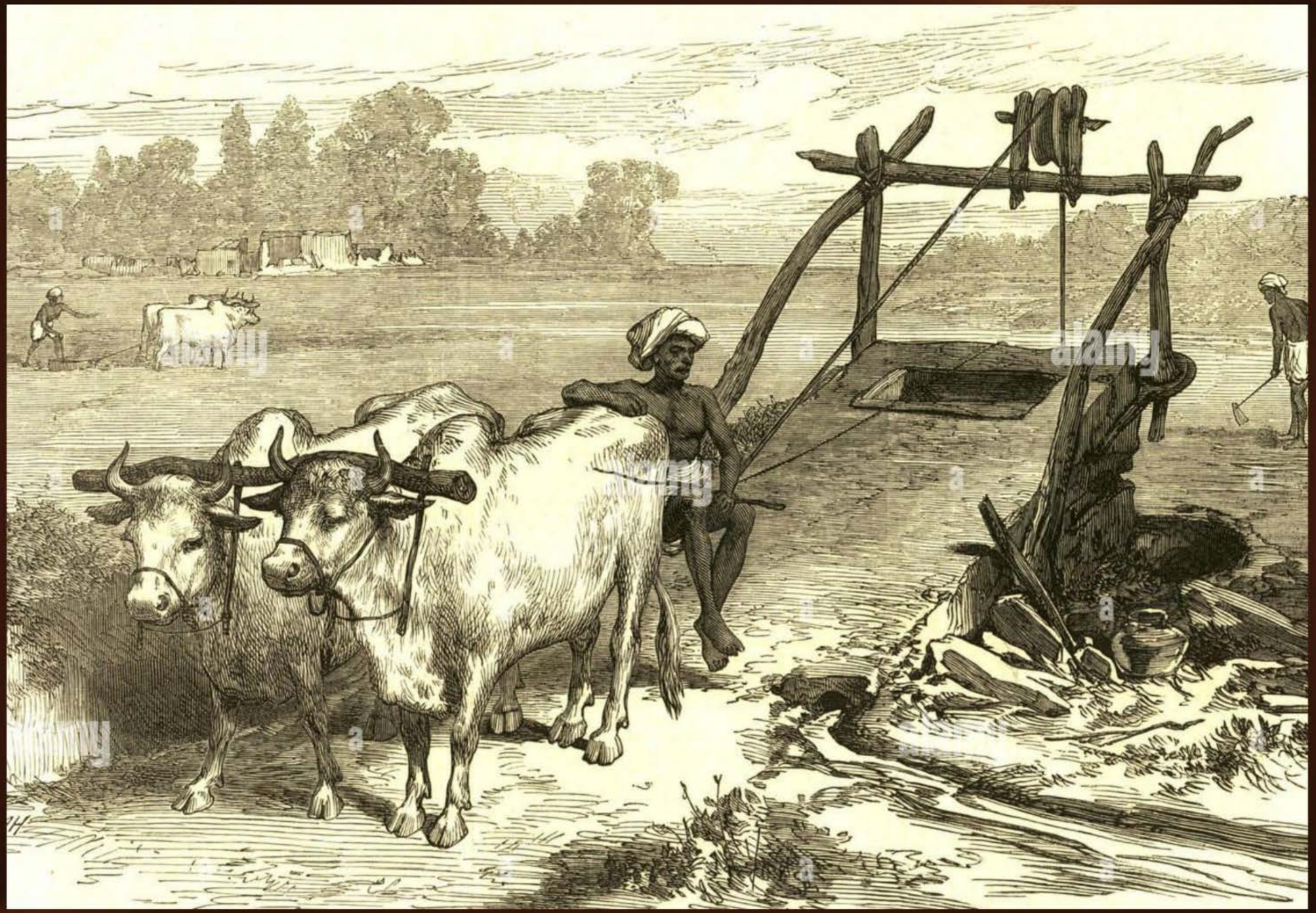
- The lift consists of a strong log laid across a fulcrum, a bucket attached to the long arm by means of a bamboo or a rope and a heavy weight attached to the other side of the log. When the bucket is full one person gently releases the rope and the bucket comes up as the stone goes down. The bucket is emptied into the field channel and again lowered into the well.



Means Of Irrigation

Mhote
Moat

(iii) **Inclined plane method:** This method is also known as mhote. In this type, a pair of bullocks move down from the slope (inclined plane), specially constructed from the wall of the well, a bucket or a leather bag which is discharged into the connecting channel pulling from behind. The rope goes through pulley in this process. After discharging the water, the bullocks walk up the slope until they reach the top by which time the bucket reaches the surface of water and gets filled. The process is repeated again and again to irrigate the fields.




alamy
J. Macfarlane

Advantages of Well Irrigation

(i) Wells can be dug at a very low cost which is well within the means of poor farmers.

(ii) Oxen which are kept for ploughing the land can be utilised for drawing water from the wells at no extra cost to the farmers.

(iii) By the use of pumps and tubewells, water can be lifted even from great depths.

Disadvantages of Well Irrigation

(i) Wells depend on underground water resources whose distribution varies from region to region.

ii) It is difficult to dig wells or bore tubewells in the hilly regions of the north and stony areas of the Peninsula.

(iii) The traditional wells dry up due to the over-withdrawal of the groundwater and lowering of the water table.

(iv) Availability of electricity and diesel to operate tubewells causes many problems and is costly for farmers.

Tubewells

A tubewell is a deeper well at a depth of 20-30m from which water is lifted up with the help of power driven pumps. It is also known as lift method'.

The ideal conditions for the drilling of tubewells are:

- availability of plentiful water at great depths,
- level land and soft soil for the tubewells to be set up,
- large fertile area in its surrounding area to be irrigated; and
- availability of regular cheap electricity to run the tubewells.



Tubewells

Tubewells are used in the northern plains in the States of Punjab, Haryana, Uttar Pradesh, Bihar, West Bengal, Rajasthan, Madhya Pradesh and Gujarat.

Advantages of Tubewell Irrigation

Perennial
Year long

(i) It is an independent source of irrigation.
(ii) It brings up clean water.
(iii) It can irrigate large area of agricultural land.
(iv) It is easier to bring up large amount of water in a short time period.
(v) It is a perennial means of irrigation as it is drilled up to permanent water table. It can be switched off when not required.

Disadvantages of Tubewell Irrigation

(i) Excessive use of tubewell water can lead to depletion of groundwater.

(ii) It is useless if the water is brackish. → Salty

(iii) It is expensive as it requires continuous supply of electricity.

Canals → नहर

India has one of the world's largest canal systems stretching over more than one lakh kilometres. The irrigation canals are of two types:

(i) **Inundation Canals:** These canals are taken out directly from the rivers without constructing any barrage or dam. Such canals use the excess water of rivers at the time of floods and remain operational during rainy season. They get supply of water only when the rivers are in flood. Therefore, these canals have limited use.







Canals

(ii) **Perennial Canals:** These canals are taken out from the perennial rivers by constructing barrages or dams to regulate the flow of water. They can draw water throughout the year and irrigate large areas. Most of the canals in India belong to this category.



Canals

- Canal irrigation is in practice in Uttar Pradesh, Madhya Pradesh, Andhra Pradesh, Rajasthan, Haryana, Jammu and Kashmir, Assam and Tripura.
- Canals also supply water to a sizeable part of the irrigated area in Bihar, Odisha, Karnataka, Tamil Nadu and West Bengal.
- Mizoram, the least irrigated State, is solely dependent on canals. The Northern Plains have an elaborate system of Canals.

Advantages of Canal Irrigation

(i) The rainfall deficiency in some areas is made up by canals taken out from the perennial rivers.

(ii) In arid areas like Rajasthan, canals have proved to be boon as a result of which sandy patches are yielding good agricultural harvests.

Advantages of Canal Irrigation

(iii) **Canal irrigation has enabled Punjab and Haryana to be called the 'granary of the country' and nucleus of the Green Revolution.**

(~~iv~~) Major parts of Tamil Nadu enjoy rainfall during winter season while summer remains dry. Thus, irrigation is needed in summer to make up the lack of rainfall. Canal irrigation provides a solution to this problem.

Disadvantages of Canal Irrigation

(i) At some places perennial unlined canals make the once cultivable soil completely waterlogged. Complete saturation of the soil with water may give rise to swamps.

(ii) When waterlogged land is exposed to the sun, the water evaporates and draws harmful salts to the surface. Over time, this salt crust makes the soil completely barren and unfit for farming.

(iii) Due to waterlogging, the capacity of soil to absorb water decreases.

(iv) A massive amount of water is lost during transit through evaporation under the sun and seepage into the ground before it ever reaches the fields.

Important Canals in India

States	Canals
Punjab	Sirhind canal, Bhakra canal, Upper Bari Doab canal.
Uttar Pradesh	Sharda canal, Upper Ganga canal, Lower Ganga canal.
Haryana	Western Yamuna canal, Gurgaon canal, Bhakra canal.
Maharashtra	Tapi canal, Pravara canal, Warna canal.
Rajasthan	Indira Gandhi canal, Bikaner canal.
Tamil Nadu	Mettur canal, Aliyar canal, Lower Bhavani Project canal.
Bihar	Son canal, Kosi canal and Gandak canal.
Karnataka	Vishweshwarya canal, Malprabha project canal.


Saturated

Tanks

Water Reservoir or Container

- Tank refers to a small bund of earth or stones or large container or reservoir built across a stream to collect water which is then led through narrow channels to the cultivated fields.

- Tank irrigation is in practice in Peninsular India including Maharashtra and Gujarat. Tanks constitute a special feature of irrigation in the Deccan because of the following reasons:





Tanks

(i) The rivers of the Deccan are not snow-fed and are mainly dependent on the rainwater;

(ii) There are many streams which become torrential during the rainy season but dry up when the rain ceases.

(iii) The hard rocks in the area do not absorb water, so wells also cannot be made there.

(iv) The terrain of the Deccan Plateau is uneven with many natural depressions where tanks can be easily built.

Tanks

- Tank irrigation has reached its highest perfection in the south, specially in Andhra Pradesh, Telangana and Tamil Nadu.
- In Tamil Nadu, it is significant in the districts of Chingleput, North Arcot and South Arcot; in Nellore in Andhra Pradesh and in Warangal in Telangana.
- States of West Bengal and Rajasthan too have some irrigation tanks, particularly in their southern and south-eastern regions respectively.
- Some irrigation tanks do exist in Bundelkhand area in Uttar Pradesh and South Bihar.

Advantages of Tank Irrigation

(i) In the uneven rocky plateau of Peninsular India, where rainfall is highly seasonal, tank irrigation is useful.

(ii) In the large stretches of the Indian plateau, due to hard rocky terrain, wells and canals are difficult to construct, so only tank irrigation is feasible.

(iii) Tank irrigation is highly significant in storing the abundant rainwater that would otherwise flow out and go waste.

Disadvantages of Tank Irrigation

Siltation
↳ Layering of rock or soil on the surface

(i) Tanks get silted up soon and regular desilting is required to make them suitable for irrigation. ↳ clean

(ii) When the monsoons fail these tanks go dry and therefore, are not dependable source of irrigation.

(iii) Most of the tanks are non-perennial and become dry during the winter and summer seasons when water is urgently required for rabi and zaid crops.

Disadvantages of Tank Irrigation

(iv) Due to large area coverage and its shallow depth, huge quantity of stored water either evaporates or sinks underground without being used in the irrigation.

(v) Tanks occupy a large fertile area which otherwise could have been utilised for growing crops.

(vi) The lifting of water from the tanks and bringing it to the fields is a hard and costly task.

Drawbacks of Conventional Methods of Irrigation

The conventional irrigation systems suffer from a number of constraints leading to improper utilisation of water. These are the following:

(i) In the conventional system of irrigation a large quantity of water is not properly utilised.

Drawbacks of Conventional Methods of Irrigation

(ii) Crops are usually subjected to cyclic changes of flooding and water stress situations, by providing heavy irrigation at one time and leaving the fields to dry up for about 10 to 15 days. The moisture availability to the crops therefore keeps on changing. This results in poor yields in the crops.

(iii) The fields situated in low areas always get excess water causing prolonged waterlogging. Thus, crops are subjected to waterlogging resulting in poor yields.

Drawbacks of Conventional Methods of Irrigation

(iv) In the fields about 10-15per cent of land is utilised for preparing channels and distributions, etc. which decreases effective area of cultivation.

(v) Extensive areas of land in the arid and semi-arid regions of India are no longer used for cultivation due to accumulation of salts.

Modern Irrigation Methods

All these constraints have led to the utilisation of modern irrigation methods which are highly scientific.

Sprinkler Irrigation: In this method of irrigation a pipe fitted with a nozzle on the top is used through which water is sprayed directly to the required plants. It is often referred as 'overhead irrigation' method.

Natural Rain

Advantages of Sprinkler Irrigation

(i) There is no loss of water by seepage.

(ii) There is no loss of water by evaporation.

Disadvantages of Sprinkler Irrigation

(i) It is an expensive method of irrigation.

(ii) It is helpful for a small area.

(iii) It is used for crops which require less water.




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Minimum water loss

Holes

Drip Irrigation: It is the most advanced and efficient method of irrigation. Usually used to irrigate fruits and vegetables, this system consists of perforated pipes that are placed between rows of crops or buried along their root lines and give water directly on to the crops. As a result, evaporation is drastically reduced and irrigation water is conserved. Drip irrigation allows the grower to customise an irrigation programme most beneficial to each crop.

Drip irrigation is a highly efficient watering method where water is delivered slowly and directly to the roots of plants, drop by drop, through a network of narrow pipes and emitters. This system drastically reduces water waste by minimizing evaporation and runoff.





Which of the following *prevents wastage of water*?

(a) Canal

(b) Well

(c) Sprinkler

(d) Tube well

(iii) Which means of irrigation can be developed with minimum expenditure?

(a) Tube well

(b) Canal

(c) Well

(d) Drip irrigation

Why is there a need to conserve water in India? *Ev* [2022 S-I]

(a) A lot of water is polluted

(b) No water is found below the land in India

(c) Water is in abundance in India

(d) We receive lot of rain in summer

1. Which of the following is the most compelling geographical reason for the widespread use of tank irrigation in Peninsular India?

- A) The region receives rainfall throughout the year.
- B) The soft alluvial soil makes it easy to dig tanks.
- C) The underlying hard, impermeable rocks prevent water from seeping into the ground.
- D) There is a lack of rivers in the entire region.

8. Which of the following methods of irrigation is considered the most efficient for minimizing evaporation and water loss?

- A) Canal irrigation
- B) Sprinkler irrigation
- C) Drip irrigation
- D) Tank irrigation

(i) Mention one advantage and one disadvantage of using tank irrigation. [2]

(ii) (a) What is the advantage of using Drip irrigation? [2]

(b) Why is Drip irrigation not practiced on a large scale in India?

(iv) With reference to the picture below, answer the questions that follow: [3]



(a) Name the water conservation practice shown in the above picture.

(b) How does this system help to reduce floods?

(c) Why is water conservation necessary? (*Any two reasons*)

Thank You